Predicting Hit Songs Using Repeated Chorus

Replication Attempt — Six-Model Evaluation (ML mini project)

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Abstract

This report documents an updated replication attempt inspired by "Predicting Hit Songs Using Repeated Chorus." The supplied dataset contains 756 songs and 518 numerical audio features. After PCA-based feature selection, the notebook evaluates six classifiers using a single 80/20 train–test split: Logistic Regression, Linear Discriminant Analysis (LDA), Support Vector Machine (SVM with an RBF kernel), Random Forest, Gradient Boosting, and a Multi-Layer Perceptron (MLP) neural network. In the saved six-model results, SVM achieved the highest test accuracy (59.21%), while the MLP recorded the highest test F1 score (0.630). Random Forest and Gradient Boosting had very high training scores but lower test scores, suggesting overfitting.

1. Introduction and objective

The original study investigated whether audio descriptors from repeated-chorus segments could help predict whether a song became a hit. This experiment uses the available feature dataset and notebook to compare classification algorithms for the same hit/non-hit prediction task. The objective is to run the pipeline, compare accuracy, F1 score, precision and recall, and document limitations that affect how confidently the results can be interpreted.

2. Dataset and features

The notebook loads feature_extract_04.csv. It reports 756 observations and 523 columns before dropping the song_path and extracted_chorus_path fields. Artist and Title are excluded from modelling, leaving 518 numerical audio-feature columns. Label is the target, with 1 representing a hit and 0 a non-hit. The dataset differs from the 554-song dataset described in the reference paper.

Dataset rows	756
Numerical audio features before selection	518
PCA-selected original features	168
Training / test observations	604 / 152
Train–test split	80% / 20%
Random state	2022

3. Methodology

3.1 PCA-based feature selection

The notebook uses the Python pca package to explore principal components and explained variance. It then uses the package's topfeat output to select 168 original features marked as "best." The classifiers are trained on these selected original features; they are not trained on a 142-component PCA projection.

3.2 Train–test split and classifiers

The selected features are split using train_test_split(test_size=0.2, random_state=2022), without stratification. The six evaluated classifiers are Logistic Regression (max_iter=2000), LDA, SVM with an RBF kernel, Random Forest (100 trees), Gradient Boosting, and MLP (max_iter=500). Model settings are those in the submitted notebook.

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Accuracy is the fraction of correct predictions. Precision measures the fraction of predicted hits that are hits; recall measures the fraction of actual hits detected; F1 is the harmonic mean of precision and recall. The positive class is Label = 1.

4. Results and discussion

The table below transcribes the six-model results displayed in the uploaded notebook's results cell. Scores are rounded to three decimal places.

Model	Train acc.	Test acc.	Test F1	Test precision	Test recall
Logistic Regression	0.637	0.533	0.553	0.530	0.579
LDA	0.649	0.559	0.573	0.556	0.592
SVM (RBF)	0.528	0.592	0.587	0.595	0.579
Random Forest	1.000	0.572	0.564	0.575	0.553
Gradient Boosting	0.997	0.493	0.490	0.493	0.487
Neural Network (MLP)	0.535	0.513	0.630	0.508	0.829

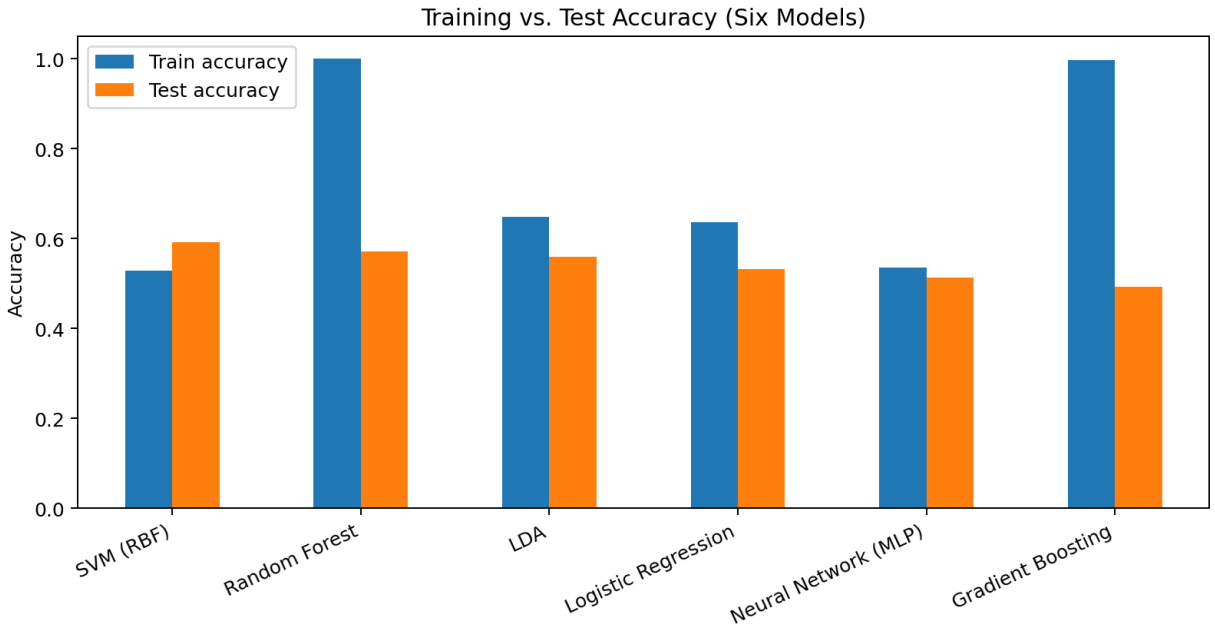


Figure 1. Training and test accuracy for the six classifiers.

4.1 Main observations

- SVM achieved the highest test accuracy: 0.592 (59.2%), with test F1 of 0.587.
- The MLP neural network achieved the highest test F1 score: 0.630, and the highest test recall: 0.829. Its test accuracy was 0.513, so the high recall should be interpreted alongside its lower precision (0.508).
- Random Forest achieved 0.572 test accuracy but 1.000 training accuracy, indicating a substantial generalisation gap and likely overfitting.
- Gradient Boosting also showed a large gap: 0.997 training accuracy versus 0.493 test accuracy.
- LDA obtained 0.559 test accuracy and 0.573 test F1. Logistic Regression obtained 0.533 test accuracy.
- These rankings apply only to this split and these settings; one held-out split is not enough to establish that one model is consistently best.

5. Limitations and reproducibility

- This is a partial replication. The supplied dataset contains 756 songs rather than the 554 reported in the reference paper. The dataset we used is a newer version and has new artists and songs.
- The original paper evaluated a broader model family and used five-fold cross-validation; this notebook currently evaluates six models on one 80/20 split.

6. Conclusion and future work

The updated notebook successfully evaluates six classifiers using the supplied chorus-feature dataset. SVM has the best test accuracy (59.2%), while the MLP has the best test F1 score (0.630) and recall (0.829). The training–test gaps for Random Forest and Gradient Boosting suggest overfitting. The next steps are to synchronize stale notebook outputs, prevent feature-selection leakage by fitting preprocessing on training folds only, use stratified cross-validation, tune hyperparameters, and compare results with the original dataset and protocol if available.

References and source materials

[1] “Predicting Hit Songs Using Repeated Chorus,” CS229 project report (2021), supplied reference paper. The current report uses the supplied notebook and its saved six-model output.